

Quang Dao

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🌐 <https://quangvdao.github.io/>

🔍 Google Scholar

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Education

- 2022–Present **Carnegie Mellon University, Pittsburgh, PA**
PhD in Computer Science. *Advisors:* Aayush Jain and Riad Wahby.
- 2020–2022 **University of Michigan, Ann Arbor, MI**
MA in Mathematics. *Advisor:* Paul Grubbs
- 2016–2020 **Columbia University, New York, NY**
BA in Mathematics and Computer Science

Research Interests

My research focuses on the **design**, **formal verification**, and **performance optimization** of **succinct zero-knowledge proofs (SNARKs)**. I founded and co-lead the development of **ArkLib**, a Lean library for compositionally verifying SNARK constructions, and am also a co-lead developer of **VCV-io**, a Lean framework for reasoning about cryptographic protocols that ArkLib builds on. I contribute to **Jolt**, a state-of-the-art zero-knowledge virtual machine (zkVM). My earlier work also explored foundational post-quantum cryptography from code-based assumptions.

Publications

(* = first author(s))

8. Behzad Abdolmaleki, Matteo Campanelli, **Quang Dao**, Hamidreza Khoshakhlagh. On the Simulation-Extractability of Proof-Carrying Data. *PKC 2026*.
7. Carl Kwan*, **Quang Dao***, Justin Thaler. Verifying Jolt zkVM Lookup Semantics. *Financial Cryptography 2025*.
6. **Quang Dao**, Aayush Jain. Lossy Cryptography from Code-Based Assumptions. *CRYPTO 2024*. **Best Paper from Early Career Researchers**.
5. **Quang Dao**, Aayush Jain, Zhengzhong Jin. Non-Interactive Zero-Knowledge from LPN and MQ. *CRYPTO 2024*.
4. **Quang Dao**, Yuval Ishai, Aayush Jain, Huijia Lin. Multi-party Homomorphic Secret Sharing and Sublinear MPC from Sparse LPN. *CRYPTO 2023*.
3. **Quang Dao***, Jim Miller*, Opal Wright, Paul Grubbs. Weak Fiat-Shamir Attacks on Modern Proof Systems. *IEEE S&P 2023*. **Distinguished Paper Award**.
2. **Quang Dao**, Paul Grubbs. Spartan and Bulletproofs are simulation-extractable (for free!). *EUROCRYPT 2023*.
1. **Quang Dao**, Julian Wellman, Calvin Yost-Wolff, Sylvester W. Zhang. Rowmotion Orbits of Trapezoid Posets. *The Electronic Journal of Combinatorics*, P2-29, 2022.

Preprints

4. Devon Tuma*, **Quang Dao***, James Waters, Alexander Hicks, Nicholas Hopper. VCVio: Verified Cryptography in Lean via Oracle Effects and Handlers. *ePrint 2026*.
3. **Quang Dao***, Ari Biswas, Liam Eagen, Andrew Milson, Shahar Papini, Justin Thaler. The Sum-Check Protocol over the Monomial Basis, and Other Optimizations. *ePrint 2026*.
2. **Quang Dao***, Zachary DeStefano*, Suyash Bagad, Yuval Domb, Justin Thaler. Speeding Up Sum-Check Proving (Extended Version). *ePrint 2026*. (*Combines and improves upon results from ePrint 2025/1117, 2024/1046, and 2024/1210.*)
1. **Quang Dao**, Justin Thaler. Constraint-Packing and the Sum-Check Protocol over Binary Tower Fields. *ePrint 2024*.

Projects

- ArkLib** github.com/Verified-zkEVM/ArkLib. Lean library for formally verified SNARKs. Founder and co-lead developer.
- VCV-io** github.com/Verified-zkEVM/VCV-io. Lean framework for reasoning about cryptographic protocols; ArkLib is built on top of it. Co-lead developer.
- Jolt** github.com/a16z/jolt. State-of-the-art zero-knowledge virtual machine (zkVM) based on structured lookup arguments. Main contributor.

Grants & Fellowships

- 2025 **Ethereum Foundation Grant**
Grant for the formalization of the FRI-Binius Polynomial Commitment Scheme
Amount: \$24,000
- 2025 **Ethereum Foundation Grant**
Grant for the development of ArkLib, a Lean library for formally verifying SNARKs
Amount: \$100,000
- 2024-2025 **Quad Fellowship**
One-year fellowship for US-based PhD students in STEM fields
Amount: \$40,000
- 2024-2025 **CyLab Fellowship**
One-year fellowship from CyLab at Carnegie Mellon University
Amount: \$50,000

Honors & Awards

- 2024 **Best Paper from Early Career Researchers**
Awarded to the best paper from early career researchers at CRYPTO 2024
- 2023 **Distinguished Paper Award**
Awarded to top 6% of accepted papers at IEEE Security & Privacy 2023
- 2020 **Russell C. Mills Award**
Awarded to 2 seniors for excellence in computer science at Columbia
- 2017 - 2019 **Van Amringe Math Prize**
Awarded annually to the top 3 non-senior students in math at Columbia
- 2016, 2018 **Putnam Math Competition**. Honorable Mention (top 50)
- 2016 **International Math Olympiad**. Silver Medal

Industry Experience

- Jan 2026 – Present Part-Time Researcher at LayerZero.
- Summer 2025 Part-Time Contractor with a16z crypto.
- Summer 2024 Research Intern at a16z crypto.

Talks

9. Evolving the Foundations of ArkLib
 - ZKProof 8, Rome (May 2026)
8. ArkLib: Compositional Verification of SNARKs
 - Real World Crypto (Mar 2026)
 - Verified zk(E)VM Seminar (Jan 2026)
 - King's College London (Oct 2025)
 - University of Cambridge (Oct 2025)
 - Ethereum's PQ Interop (Oct 2025)
 - CMU PLunch (Sep 2025)
 - University of Luxembourg Crypto Seminar (July 2025)
 - CRYPTO Day, Telecom Paris (June 2025)
 - EPFL Crypto Seminar (June 2025)
 - CMU Formal Cookies (Oct 2024)
7. Lossy Cryptography from Code-Based Assumptions
 - MIT CIS Seminar (Nov 2024)
 - BUSEC Seminar (Nov 2024)
 - NYU Crypto Reading Group (Nov 2024)
 - CMU CyLab Crypto Seminar (Sep 2024)
 - UCLA Crypto Reading Group (Apr 2024)
 - UToronto Theory Seminar (Mar 2024)
6. Non-Interactive Zero-Knowledge from LPN and MQ
 - CMU CyLab Crypto Seminar (Sep 2024)
5. Advanced Security for SNARKs: A Survey
 - a16z Crypto Research Seminar (Jun 2024)
4. Multi-party Homomorphic Secret Sharing and Sublinear MPC from Sparse LPN
 - JP Morgan AlgoCRYPT Seminar (Dec 2023)
 - CMU Crypto Seminar (Nov 2023)
 - NTT Research Seminar (Oct 2023)
 - CyLab Partners Conference (Oct 2023)
 - Vietnam Mathematical Congress (Aug 2023)
3. Weak Fiat-Shamir Attacks on Modern Proof Systems
 - Real World Crypto (Mar 2024)
 - CMU CyLab Security Seminar (Nov 2023)
 - Cornell Security Seminar (Sep 2023)
 - NYU Crypto Reading Group (Sep 2023)
 - Workshop on Attacks in Cryptography (Aug 2023)
 - IEEE S&P (May 2023)

2. Spartan and Bulletproofs are simulation-extractable (for free!)
 - Stanford Crypto Reading Group (May 2023)
 - Telecom Paris Seminar (May 2023)
 - Lattices Meet Hashes Workshop, EPFL (May 2023)
 - CMU Crypto Seminar (April 2023)

Service

Program Committee	ASIACRYPT 2026
External Reviewer	2026: CRYPTO
	2025: EUROCRYPT, IEEE S&P, CRYPTO, RANDOM, TCC
	2024: STOC, EUROCRYPT, TCC
	2023: ASIACRYPT, TCC, FOCS
	2022: CRYPTO, EUROCRYPT
Co-Organizer	2022-2025: CMU Crypto Seminar

Teaching

2023-2025	Western Pennsylvania ARML Team , <i>Assistant Coach</i> , CMU
Fall 2023	Undergraduate Quantum Computation , <i>Teaching Assistant</i> , CMU
2020–2021	Calculus I , <i>Lead Instructor</i> , University of Michigan
2017, 2021	Math & Science Summer Program (MASSP) , <i>Mentor</i> , Vietnam

Mentorship

Chung Nguyen	Post-baccalaureate, 2025–Present
James Waters	Undergraduate, Fall 2025–Summer 2026
Liam Schilling	Undergraduate, Spring & Fall 2025
Ian Martin	Undergraduate, Summer 2025
Jack Zhu	Undergraduate, Spring 2025
Christian Ang	Undergraduate, Spring 2025

Miscellaneous

Languages	English (fluent), Vietnamese (native), French (elementary), Malayalam (elementary)
Other writing	Quang Dao , Christina Meng, Julian Wellman, Zixuan Xu, Calvin Yost-Wolff, Teresa Yu. Extended Nestohedra and their Face Numbers. <i>arXiv</i> 2019.